第八讲 弯曲内力

作业：5-2,4(a,b),5,6;

第五章 弯曲内力

5.1 梁的约束与类型

5.2 剪力弯矩图

5.3 剪力弯矩与载荷集度间的微分关系

5.4 刚架和曲杆的内力

第五章 弯曲内力

5.1 梁的约束与类型

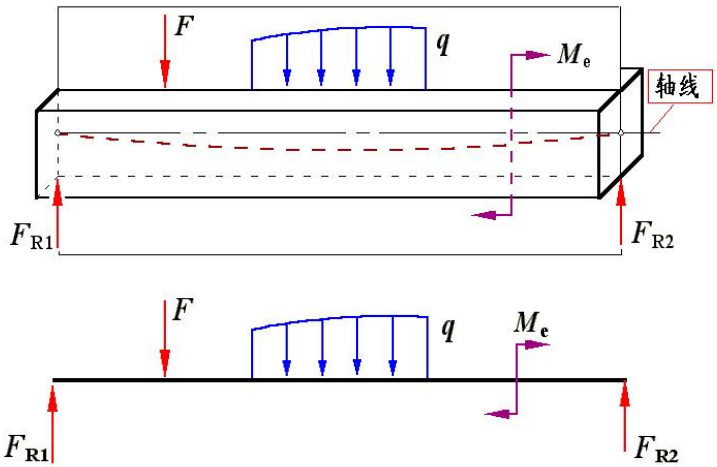
1、弯曲实例

2、约束形式与反力

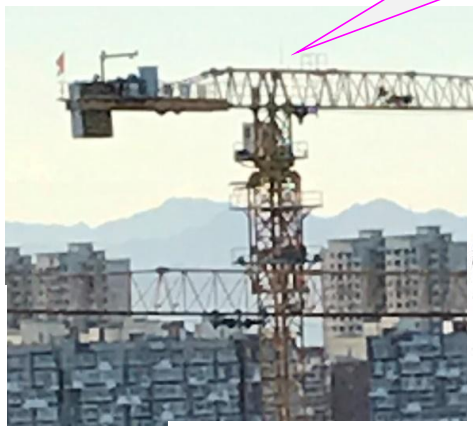
3、梁的类型

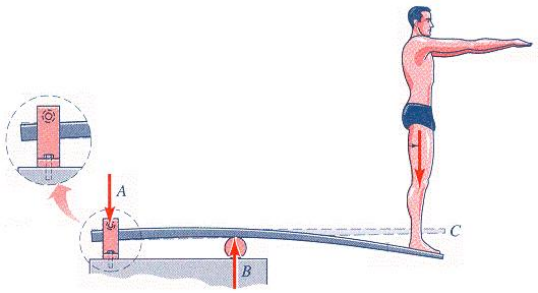
1、弯曲实例

外力：横向力和轴线平面内的力矩

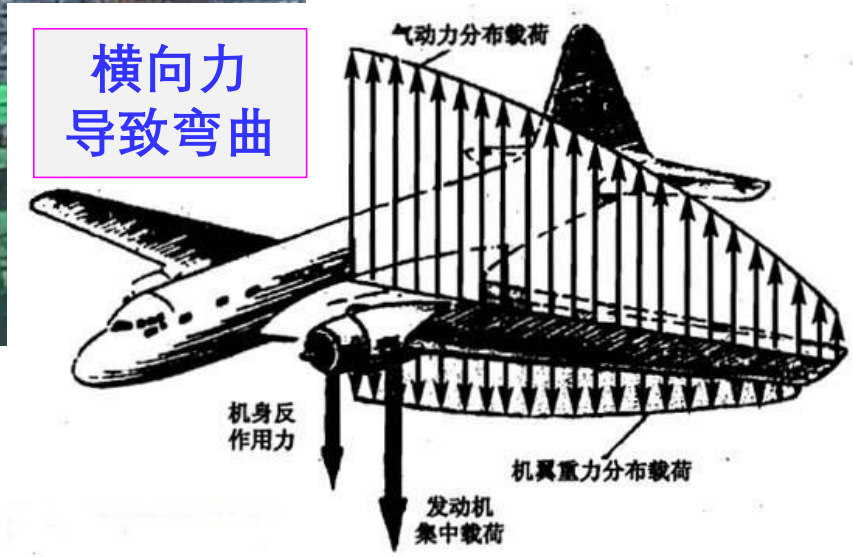


集中力矩作用





横向力
导致弯曲

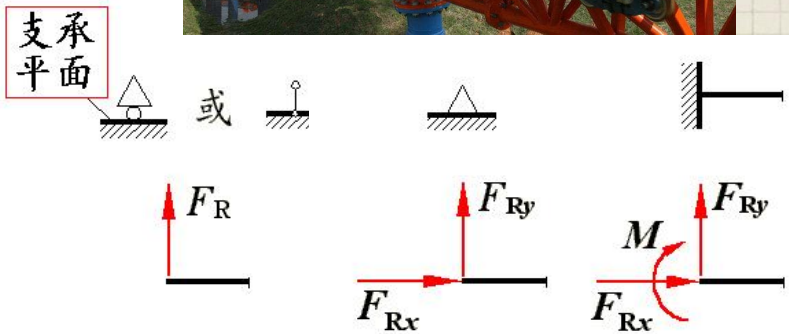


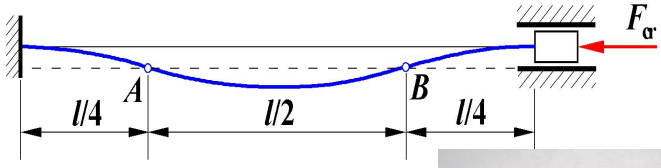
弯曲：轴线变弯（曲率改变）；

梁：以弯曲为主的杆件，

以轴线表示。

2、约束形式与反力





滑动铰支座：垂直于支承平面的支反力 F_R；

固定铰支座：支反力 F_{Rx} 与 F_{Ry}；

固定（悬臂）端：支反力 F_{Rx}，F_{Ry} 与矩为 M 的支反力偶矩；

定向支座：支反力 F_R 与反力矩 M（相当于“可动”固定端）；

梁间铰链：反力 F_{Rx} 与 F_{Ry}

3、梁的类型

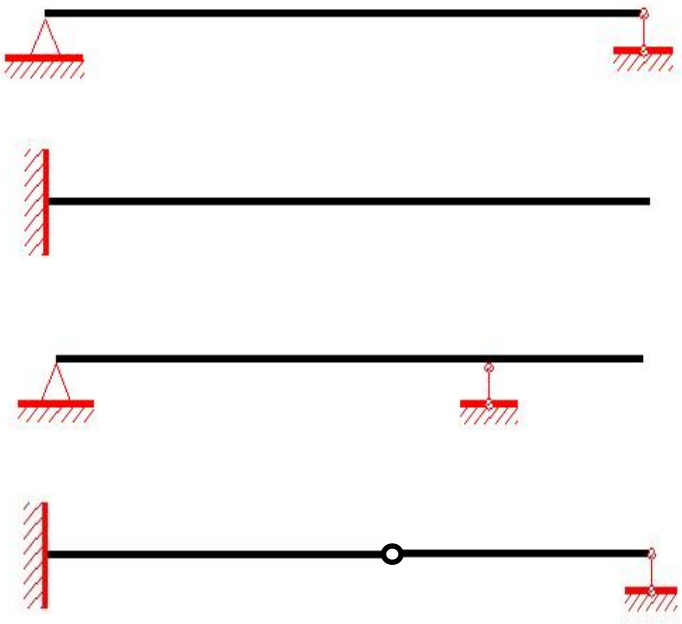
常见静定梁（平面内）
——3个约束、3个反力

简支梁：一端固定铰支、一端滑动铰支；

悬臂梁（固支梁）：一端固定、一端自由；

外伸梁：有外伸段的简支梁；

组合梁：以梁间铰链连接的梁。



静不定梁　约束反力数超过有效平衡方程数的梁

第五章 弯曲内力

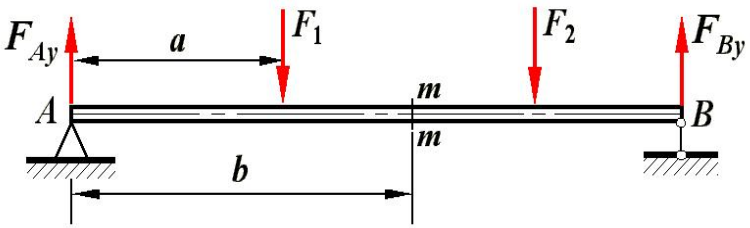
5.2 剪力弯矩图

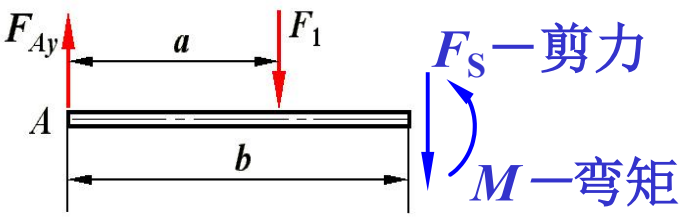
1、剪力与弯矩

2、剪力弯矩图

1、剪力与弯矩

截面法求剪力弯矩





分离体平衡

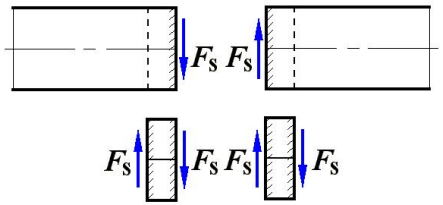
$$F_S = F_{Ay} - F_1$$

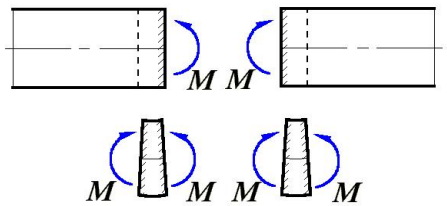
$$M = F_{Ay}b - F_1(b-a)$$

剪力—横截面内的力

弯矩—轴线平面内的力矩

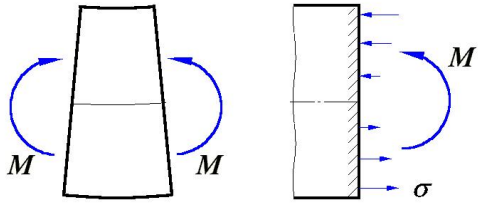
剪力弯矩的符号规定





微段顺时针转动
剪力为正

微段弯曲呈凹形
弯矩为正



微段顶部受压
弯矩为正

2、剪力弯矩图

很重要　挺简单
守规矩　会检查

关于静力等效

剪力弯矩方程

$F_S = F_S(x)$ —剪力方程

$M = M(x)$ —弯矩方程

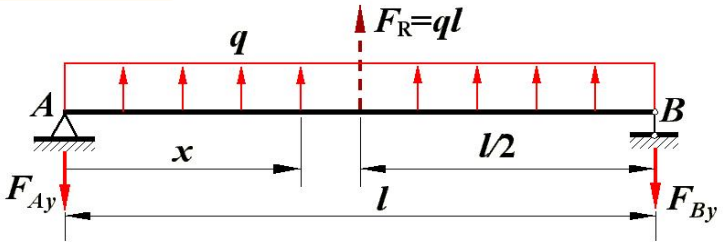
表示剪力弯矩
与截面位置的关系

剪力弯矩图（QM图）

危险截面

强度计算

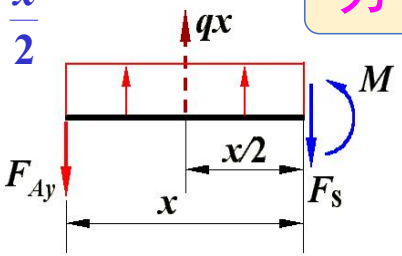
$F_{Ay} = F_{By} = \dfrac{ql}{2}$



$F_S = -F_{Ay} + qx$

$M = -F_{Ay} \cdot x + qx \cdot \dfrac{x}{2}$

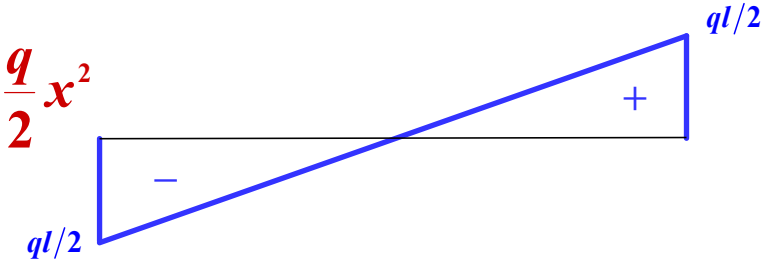
分离体平衡



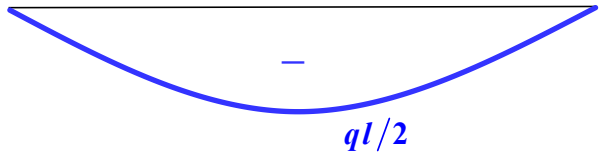
剪力弯矩设正

$F_S = -\dfrac{ql}{2} + qx$

$M = -\dfrac{ql}{2}x + \dfrac{q}{2}x^2$



$x = \dfrac{l}{2}, \quad |M_{max}| = \dfrac{ql^2}{8}$



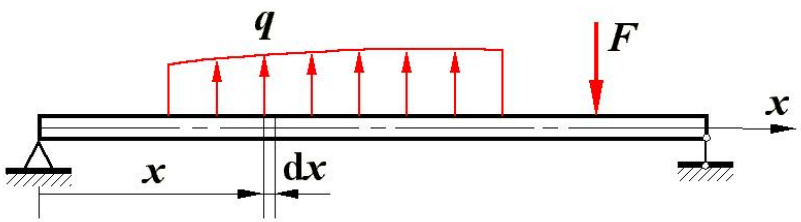
第五章 弯曲内力

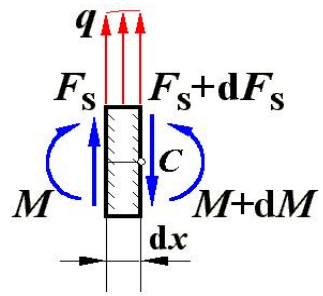
5.3 剪力弯矩与载荷集度间的微分关系

1、微分关系——梁段平衡方程

2、微分关系的几何意义及其应用

1、微分关系——梁段平衡方程





$$\sum F_y = 0,\quad F_S + q\mathrm{d}x - (F_S + \mathrm{d}F_S) = 0 \quad \Rightarrow \quad \boxed{\mathrm{d}F_S = q\mathrm{d}x}$$

$$\sum M_C = 0,\quad M + \mathrm{d}M - q\mathrm{d}x \cdot \frac{\mathrm{d}x}{2} - F_S\mathrm{d}x - M = 0 \quad \Rightarrow \quad \boxed{\mathrm{d}M = F_S\mathrm{d}x}$$

$$\frac{\mathrm{d}F_S(x)}{\mathrm{d}x} = q(x) \qquad \frac{\mathrm{d}M(x)}{\mathrm{d}x} = F_S(x)$$

$$\frac{\mathrm{d}^2M(x)}{\mathrm{d}x^2} = q(x)$$

梁段平衡方程

分布载荷引起剪力的增量；

剪力引起弯矩的增量。

q 向上为正，x 向右为正

1、微分关系——梁段平衡方程

集中力作用处

$$F_R = F + q\,\mathrm{d}x + F_L$$

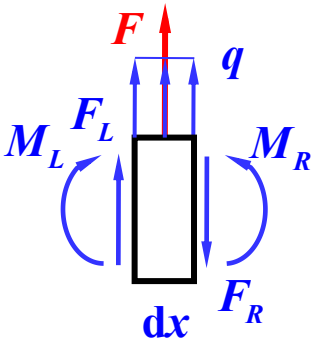
$$M_R = F\frac{\mathrm{d}x}{2} + \frac{q(\mathrm{d}x)^2}{2} + M_L + F_L\,\mathrm{d}x$$

$$F_R - F_L = F$$

$$M_R = M_L$$

剪力突变

弯矩连续



集中力矩作用处

$$F_R = q\,\mathrm{d}x + F_L$$

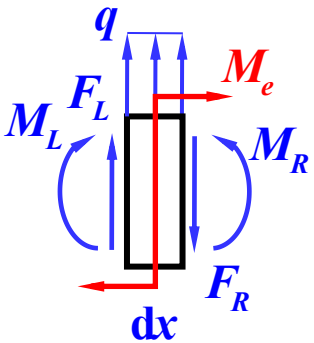
$$M_R = F_L\,\mathrm{d}x + \frac{q(\mathrm{d}x)^2}{2} + M_e + M_L$$

$$F_R = F_L$$

$$M_R - M_L = M_e$$

弯矩突变

剪力连续



- 对于杆件拉压、扭转问题，内力与载荷有类似的微分关系；
- 其力学本质，就是杆件微段的平衡方程。

2、微分关系的几何意义及其应用

$$\frac{\mathrm{d}F_S}{\mathrm{d}x}=q,\quad \frac{\mathrm{d}M}{\mathrm{d}x}=F_S,\quad \frac{\mathrm{d}^2M}{\mathrm{d}x^2}=q$$

q 向上为正
x 向右为正

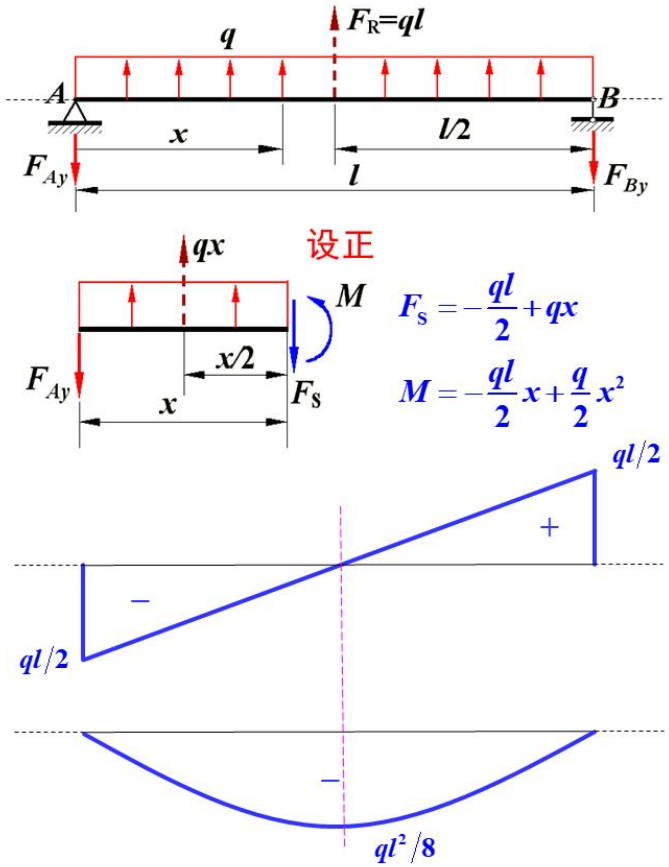
很重要　挺简单
守规矩　会检查

q	$q(x)=0$	$q(x)=c<0$	$q(x)=c>0$
F_S图			
M图	直线	2次凸曲线	2次凹曲线

仅表示凸凹，不确定有极值

F_S	>0	$=0$	<0
M	上升	极值	下降

集中力处，F_S图突变
集中力矩处，M图突变



本讲：剪力弯矩图

很重要，挺简单

守规矩，会检查

下讲：轴向拉压应力